

REAL-VALUED JORDAN CURVES AND NONREAL LIMITING SPECTRA OF BANDED TOEPLITZ MATRICES

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ABSTRACT. We construct Laurent polynomials that are real-valued on a Jordan curve surrounding the origin but whose banded Toeplitz matrices have nonreal limiting spectrum. This refutes the weakest surviving form of a conjecture of Shapiro and Štampach.

1. INTRODUCTION AND MAIN RESULTS

Let

$$(1.1) \quad b(z) = \sum_{k=-r}^s b_k z^k, \quad r, s \geq 1, \quad b_{-r} b_s \neq 0,$$

be a Laurent polynomial, and let

$$T_n(b) = (b_{i-j})_{i,j=1}^n$$

be its finite Toeplitz sections. We are interested in the limiting spectrum

$$(1.2) \quad \Lambda(b) = \{\lambda \in \mathbb{C} : \lim_{n \rightarrow \infty} \text{dist}(\lambda, \text{spec}(T_n(b))) = 0\}.$$

For each $\lambda \in \mathbb{C}$, denote the roots of

$$(1.3) \quad z^r(b(z) - \lambda) = 0$$

by $z_1(\lambda), \dots, z_{r+s}(\lambda)$, repeated according to multiplicity and ordered so that

$$0 < |z_1(\lambda)| \leq \dots \leq |z_{r+s}(\lambda)|.$$

The following characterisation of the limiting spectrum is classical [5].

Theorem 1.1. *Let b be a Laurent polynomial as defined in 1.1. Then it holds that*

$$(1.4) \quad \begin{aligned} \Lambda(b) &= \bigcap_{R>0} \sigma(T(b(Rz))) \\ &= \{\lambda \in \mathbb{C} : |z_r(\lambda)| = |z_{r+1}(\lambda)|\} \end{aligned}$$

An important question is when the limiting spectrum is real-valued. While this is trivially the case for self-adjoint operators, it has also been numerically observed for a wide range of non-self-adjoint Toeplitz matrices.

Conjecture 1.2. Let b be a Laurent polynomial as in (1.1). The following are equivalent

- (1) $\Lambda(b) \subset \mathbb{R}$,
- (2) the set $b^{-1}(\mathbb{R})$ contains a Jordan curve with the origin in its interior,
- (3) $\sigma(T_n(b)) \subset \mathbb{R}$ for all $n \in \mathbb{N}$.

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The above conjecture was first asserted as a Theorem in [7], unfortunately the proof was later discovered to be erroneous [6] and remains the subject of active research [2, 8]. The implication $(1) \implies (2)$ remains proven as it was unaffected by the error and is also known to hold in the case of matrix-valued symbols [4]. Common belief was that the implication $(2) \implies (3)$, or at least the weaker result $(2) \implies (1)$ would still hold. The former has recently been disproven in [3], but the later direction remained open until now.

In this paper, we provide two surprisingly simple counterexamples to $(2) \implies (1)$ (and consequently also the stronger claim $(2) \implies (3)$). The main result, Theorem 1.3, shows that the open part Conjecture 1.2 is false, even when restricting to symbols with real-valued coefficients.

Theorem 1.3. *Define*

$$(1.5) \quad h(z) = z^2 + 97z + 103 - \frac{10201}{z}, \quad b(z) = h(z)^2.$$

Then b has real Laurent coefficients, with $r = 2$ and $s = 4$, and the following statements hold.

(i) *There is a real-analytic Jordan curve $\gamma \subset \mathbb{C} \setminus \{0\}$ satisfying*

$$\text{wind}(\gamma, 0) = 1, \quad b(\gamma) = [-4000000, 0].$$

(ii) *Set*

$$(1.6) \quad \rho = \frac{101}{\sqrt{97}}, \quad L = \frac{20192}{97}, \quad \lambda_0 = L^2.$$

In a neighborhood of λ_0 , the set $\Lambda(b)$ is a regular real-analytic arc transverse to the real axis. It has a parametrization $\lambda(t)$, for real t near 0, such that

$$(1.7) \quad \lambda(t) = \lambda_0 + iCt + O(t^2), \quad C = \frac{2L\rho(97^2 - \rho^2)}{97} > 0.$$

All six roots of $z^2(b(z) - \lambda)$ are simple throughout a sufficiently small neighborhood of λ_0 .

Consequently, $\Lambda(b) \not\subset \mathbb{R}$.

A second example, proved in Section 4, has $r = 1$ and $s = 2$:

$$(1.8) \quad a(z) = \frac{10201}{z} + 97z + i(103 - z^2).$$

It is real-valued on the rotated curve $i\gamma$, but

$$(1.9) \quad \frac{20192}{97}i \in \Lambda(a).$$

The remainder of the paper is structured as follows: The curve γ is constructed in Section 2. Section 3 establishes the root ordering and non-reality of the spectrum. In Section 4 the simpler counterexample (1.8) is discussed in more detail.

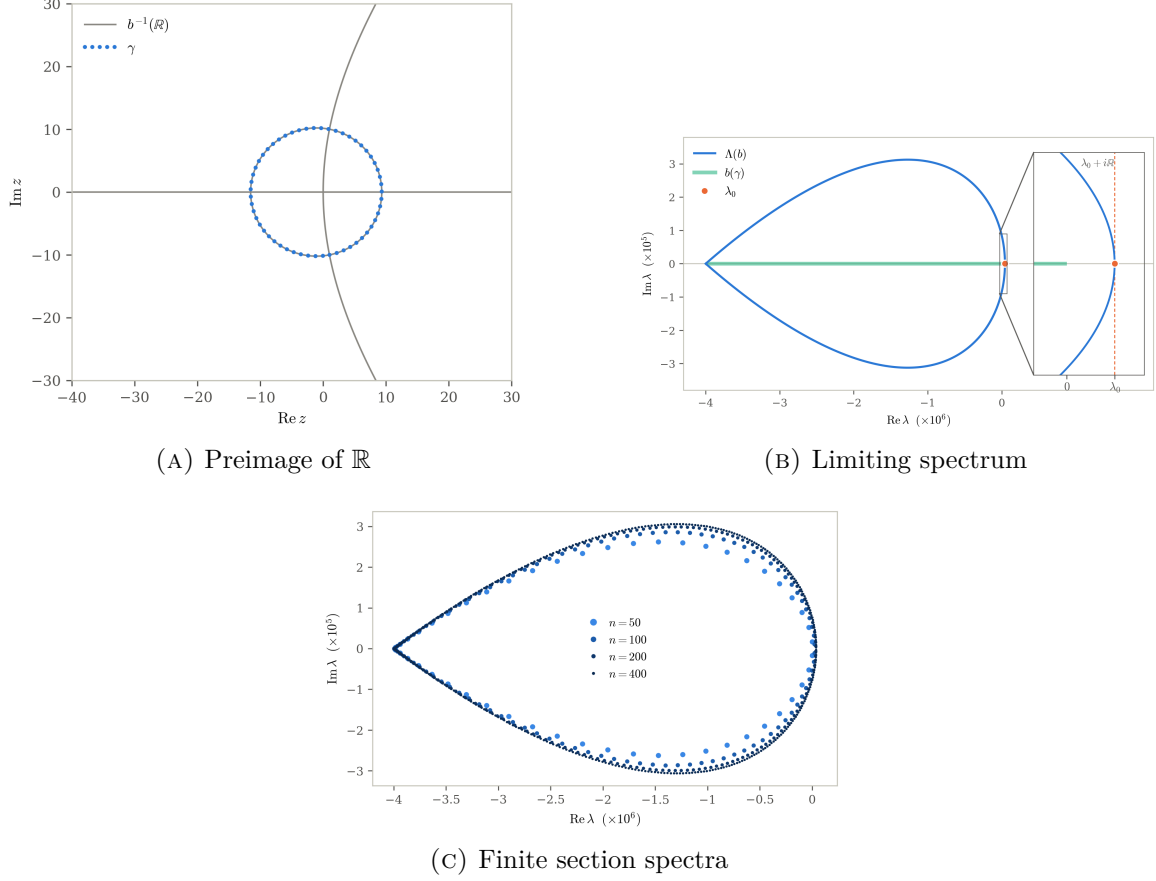
2. CONSTRUCTION OF γ

We first construct a Jordan curve on which h is purely imaginary. Squaring h will then make the symbol real on the same curve.

Lemma 2.1. *For every $t \in [-2000, 2000]$, the polynomial*

$$(2.1) \quad P_t(z) = z(h(z) - it) = z^3 + 97z^2 + (103 - it)z - 10201$$

has exactly two roots in $|z| < 30$, counted with multiplicity, and no root on $|z| = 30$.

FIGURE 1. Preimage of $\mathbb{R}$ and Limiting Spectrum for b as in Theorem 1.3

Proof. Compare P_t with

$$Q_t(z) = 100z^2 - itz - 10000.$$

The roots of Q_t are

$$\frac{it \pm \sqrt{4000000 - t^2}}{200},$$

and both have modulus 10 for $t \in [-2000, 2000]$. By factoring Q_t , we obtain on $|z| = 30$,

$$|Q_t(z)| \geq 100(30 - 10)^2 = 40000.$$

On the same circle,

$$\begin{aligned} |P_t(z) - Q_t(z)| &= |z^3 - 3z^2 + 103z - 201| \\ &\leq 30^3 + 3 \cdot 30^2 + 103 \cdot 30 + 201 \\ &= 32991 < 40000. \end{aligned}$$

Rouché's theorem gives the asserted root count, and the strict inequality excludes roots on the circle. $\square$

Proposition 2.2. *The set*

$$(2.2) \quad \gamma = \{z \in \mathbb{C} : |z| < 30, h(z) \in i[-2000, 2000]\}$$

is a real-analytic Jordan curve. Its bounded interior contains 0, and

$$h(\gamma) = i[-2000, 2000], \quad b(\gamma) = [-4000000, 0].$$

Proof. The critical points of h are obtained from

$$(2.3) \quad z^2 h'(z) = 2z^3 + 97z^2 + 10201 = (2z + 101)((z - 1)^2 + 100).$$

They are $1 + 10i$, $1 - 10i$, and $-101/2$. Their critical values are

$$(2.4) \quad h(1 + 10i) = 2000i, \quad h(1 - 10i) = -2000i, \quad h(-101/2) = -\frac{8173}{4}.$$

Each is a simple critical point: h' has a simple zero there.

For $-2000 < t < 2000$, every root of P_t is simple. Indeed, $P_t(0) \neq 0$, and a multiple root would satisfy $h(z) = it$ and $h'(z) = 0$, contrary to (2.4). By Lemma 2.1, exactly two roots lie in $|z| < 30$. They can be labeled as locally real-analytic functions

$$\alpha(t), \beta(t), \quad -2000 < t < 2000.$$

At the endpoints there are exact factorizations

$$(2.5) \quad P_{\pm 2000}(z) = (z - (1 \pm 10i))^2(z + 99 \pm 20i).$$

The third root has modulus 101. Thus the only root inside $|z| < 30$ at either endpoint is the indicated double root. Continuity of polynomial roots, together with Lemma 2.1, yields continuous extensions of both branches with

$$\begin{aligned} \alpha(-2000) &= \beta(-2000) = 1 - 10i, \\ \alpha(2000) &= \beta(2000) = 1 + 10i. \end{aligned}$$

Each extended branch is injective. For example, equality $\alpha(t_1) = \alpha(t_2)$ implies

$$it_1 = h(\alpha(t_1)) = h(\alpha(t_2)) = it_2.$$

An intersection between the two branches likewise forces equality of the parameters. At an interior parameter the two roots are distinct, so the branches have only their two endpoints in common. Their union is therefore a Jordan curve, and Lemma 2.1 shows that this union is exactly (2.2).

Away from the two endpoints, the curve is real-analytic by the implicit function theorem, since $h' \neq 0$ there. To examine an endpoint, fix one of the two signs and set $c = 1 \pm 10i$, so that $h(c) = \pm 2000i$. As c is a simple critical point,

$$h(z) - h(c) = (z - c)^2 g(z)$$

with g holomorphic near c and $g(c) \neq 0$. Choosing a holomorphic branch of $\sqrt{g}$ on a disc around c , set $\zeta(z) := (z - c)\sqrt{g(z)}$. Then $\zeta(c) = 0$, $\zeta'(c) = \sqrt{g(c)} \neq 0$, and

$$h(z) - h(c) = \zeta(z)^2,$$

so ζ maps a neighbourhood of c conformally [1, Corollary 2] onto a neighbourhood of 0, in which h becomes the map $\zeta \mapsto h(c) + \zeta^2$. Near c the curve is the preimage under h of a sufficiently short one-sided segment $h(c) \mp i[0, \varepsilon)$, with $\varepsilon > 0$. In the ζ -coordinate, this preimage is the straight-line segment

$$e^{\mp i\pi/4}(-\sqrt{\varepsilon}, \sqrt{\varepsilon}),$$

symmetric about 0, whose two halves correspond to the two root branches that coalesce at c . The curve near c is therefore the image of a straight line under the conformal map ζ^{-1} , hence a regular real-analytic arc through c .

The curve avoids 0, since $P_t(0) = -10201$ for every t . Let D be its bounded interior. If $0 \notin D$, then h would be holomorphic on a neighborhood of $\overline{D}$, with $\operatorname{Re} h = 0$ on ∂D . The maximum and minimum principles for $\operatorname{Re} h$ would give $\operatorname{Re} h = 0$ throughout D . The open mapping theorem would then force h to be constant, a contradiction. Hence $0 \in D$, and positive orientation gives $\operatorname{wind}(\gamma, 0) = 1$.

Every parameter in the segment occurs on either branch, so $h(\gamma) = i[-2000, 2000]$. Squaring proves the statement about $b(\gamma)$. $\square$

3. THE REAL-COEFFICIENT EXAMPLE

Expanding (1.5) gives

$$(3.1) \quad b(z) = z^4 + 194z^3 + 9615z^2 - 420z - 1968385 - \frac{2101406}{z} + \frac{104060401}{z^2}.$$

In particular, its pole order is 2, so the middle-root condition is $|z_2(\lambda)| = |z_3(\lambda)|$. Recall ρ, L, λ_0 from (1.6); they satisfy

$$(3.2) \quad \rho^2 = \frac{10201}{97}, \quad L = 103 + \rho^2, \quad 10 < \rho < 11, \quad 208 < L < 209.$$

3.1. Exact root ordering at a real spectral parameter.

Lemma 3.1. *All six roots of $z^2(b(z) - \lambda_0)$ are simple. Their moduli satisfy*

$$(3.3) \quad |z_1(\lambda_0)| < \rho = |z_2(\lambda_0)| = |z_3(\lambda_0)| < |z_4(\lambda_0)| < |z_5(\lambda_0)| < |z_6(\lambda_0)|.$$

The two roots of modulus ρ are ρ and $-\rho$.

Proof. Since $\lambda_0 = L^2$, the equation $b(z) = \lambda_0$ splits into $h(z) = L$ and $h(z) = -L$. For the first equation,

$$(3.4) \quad z(h(z) - L) = z^3 + 97z^2 - \rho^2z - 10201 = (z + 97)(z^2 - \rho^2).$$

Its roots are $-97, -\rho, \rho$.

For the second equation, let

$$q(z) = z(h(z) + L) = z^3 + 97z^2 + (103 + L)z - 10201.$$

The following signs are exact:

$$\begin{aligned} q(-97) &= -194L < 0, & q(-30) &= 47009 - 30L > 0, \\ q(-\rho) &= -2L\rho < 0, & q(0) &= -10201 < 0, \\ q(\rho) &= 2L\rho > 0. \end{aligned}$$

Consequently, q has a root in each of the disjoint intervals

$$(3.5) \quad (-97, -30), \quad (-30, -\rho), \quad (0, \rho).$$

These are all its roots because q is cubic. They are distinct and hence simple. The two cubic factors have no common root, since $L \neq 0$. Combining (3.4) and (3.5) proves simplicity of all six roots and the ordering (3.3). $\square$

3.2. A regular spectral arc transverse to the real axis.

Proof of Theorem 1.3. Part (i) is Proposition 2.2, and (3.1) verifies the coefficient and bandwidth assertions. It remains to prove part (ii).

By Lemma 3.1 and the holomorphic implicit function theorem, there are holomorphic root branches $u(\lambda)$ and $v(\lambda)$ near λ_0 such that

$$b(u(\lambda)) = b(v(\lambda)) = \lambda, \quad u(\lambda_0) = \rho, \quad v(\lambda_0) = -\rho.$$

The other four roots also have distinct holomorphic continuations. Shrinking the neighborhood if necessary, all six roots remain simple. The strict gaps in (3.3) ensure that exactly one of the other roots has modulus smaller than both $|u(\lambda)|$ and $|v(\lambda)|$, and the remaining three have modulus larger than both. Thus Theorem 1.1 implies in this neighbourhood

$$(3.6) \quad \lambda \in \Lambda(b) \iff |u(\lambda)| = |v(\lambda)|.$$

Define the holomorphic quotient

$$F(\lambda) = \frac{u(\lambda)}{v(\lambda)}.$$

Then $F(\lambda_0) = -1$. Since $b' = 2hh'$, equations (3.4) and (3.2) give

$$b'(\rho) = 4L(97 + \rho), \quad b'(-\rho) = 4L(97 - \rho).$$

Differentiating the inverse-branch identities yields

$$u'(\lambda_0) = \frac{1}{4L(97 + \rho)}, \quad v'(\lambda_0) = \frac{1}{4L(97 - \rho)}.$$

It follows that

$$\begin{aligned} (3.7) \quad F'(\lambda_0) &= -\frac{u'(\lambda_0) + v'(\lambda_0)}{\rho} \\ &= -\frac{97}{2L\rho(97^2 - \rho^2)} < 0. \end{aligned}$$

In particular, F is biholomorphic on a sufficiently small neighborhood U of λ_0 . We may choose U so that the intersection of $F(U)$ with the unit circle is a single open arc containing -1 . By (3.6), $\Lambda(b) \cap U$ is exactly its inverse image under F .

For real t near 0, set

$$(3.8) \quad \lambda(t) = F^{-1}(-e^{it}).$$

This is a regular real-analytic parametrization of the local spectral arc. Moreover,

$$\lambda'(0) = \frac{-i}{F'(\lambda_0)} = i \frac{2L\rho(97^2 - \rho^2)}{97} = iC.$$

This proves (1.7). Because $C > 0$, one has $\text{Im } \lambda(t) \neq 0$ for every sufficiently small nonzero real t . Hence the arc crosses the real axis transversely, and $\Lambda(b) \not\subset \mathbb{R}$. $\square$

4. A FOUR-DIAGONAL EXAMPLE WITH AN EXPLICIT NONREAL POINT

Proposition 4.1. *The Laurent polynomial*

$$a(z) = \frac{10201}{z} + 97z + i(103 - z^2)$$

is real-valued on a real-analytic Jordan curve surrounding 0, but

$$\lambda_* := \frac{20192}{97}i \in \Lambda(a).$$

At λ_ , all three roots of $z(a(z) - \lambda_*)$ are simple, and the two smallest have equal modulus.*

Proof. By direct substitution,

$$a(z) = ih(-iz).$$

Consequently, for $w \in \gamma$,

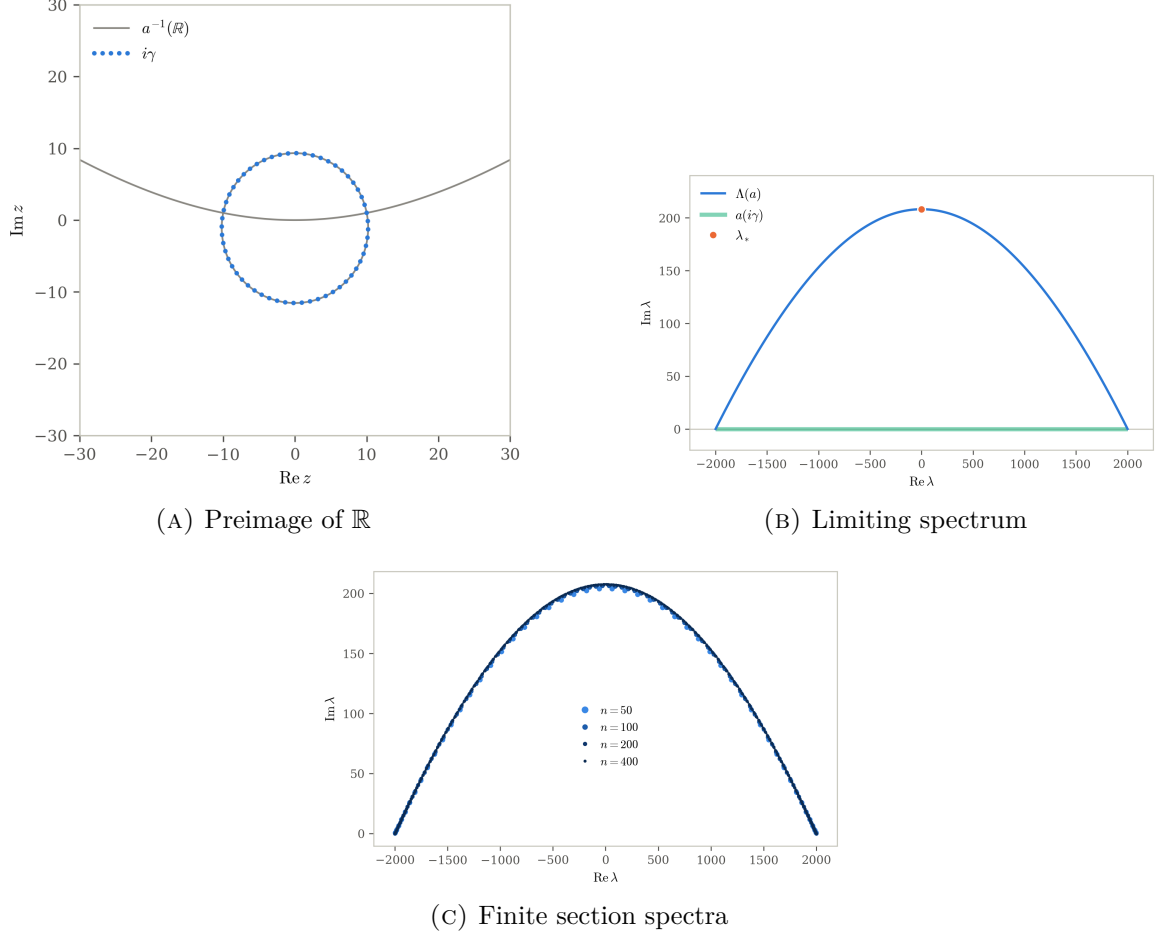
$$a(iw) = ih(w) \in \mathbb{R}.$$

Proposition 2.2 therefore shows that $i\gamma$ has the required properties; multiplication by i preserves the winding number about 0.

At $\lambda_* = iL$, the root polynomial factors as

$$\begin{aligned} (4.1) \quad z(a(z) - iL) &= -iz^3 + 97z^2 - i\rho^2z + 10201 \\ &= -i(z^2 + \rho^2)(z + 97i). \end{aligned}$$

Its roots are $i\rho, -i\rho, -97i$. They are distinct, and $\rho < 97$. Since the pole order of a is 1, the Schmidt-Spitzer characterisation of the limiting spectrum in Theorem 1.1 is the equality of the first two root moduli. Equation (4.1) gives exactly this equality at the nonreal point λ_* . $\square$

FIGURE 2. Preimage of $\mathbb{R}$ and Limiting Spectrum for a as in Proposition 4.1

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